



Null hypothesis of an ANOVA when comparing regression models?

Asked 7 years, 4 months ago Modified 6 years, 5 months ago Viewed 13k times



So I want to compare my full model with my reduced model but I am having some trouble getting my head around the hypothesis' when performing the ANOVA using R. Is the following correct:

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ANOVA(reduced model, full model)



- Null: Reduced model is as good as full model
- Alt: Full model is significantly better
- $p < 0.05 \Rightarrow$ choose full model
- $p > 0.05 \Rightarrow$ choose reduced model



ANOVA(full model, reduced model)

- Null: full model is as good as reduced model
- Alt: reduced model is significantly better
- $p < 0.05 \Rightarrow$ choose reduced model
- $p > 0.05 \Rightarrow$ choose full model

?

[r](#) [hypothesis-testing](#) [anova](#)

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asked Apr 4, 2017 at 17:22



321Chris

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I find this confusing because the question is more about R than about statistical interpretation. I really (personally, that is) couldn't care less about formatting of hypotheses in R. I do care about formatting of hypotheses for statistics. – [Carl](#) Apr 4, 2017 at 17:52



It is easy to check this: The anova function will give you the same p-value whether you put the full model first or the full model second. So the decision rule you are proposing cannot be correct. – [Sal Mangiafico](#) Dec 23, 2017 at 15:23

2 Answers

Sorted by: Highest score (default)





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Old thread, but still. To directly answer your question: it is not correct. In R, **ANOVA(reduced model, full model)** and **ANOVA(full model, reduced model)** yield the same output. Under the assumption that the models are nested (meaning that *all factors of one of the models are present in the other*), the anova function finds the factors that have been removed in the reduced model as compared to the full model, and tests whether those removed factors are different from zero. If they are significantly different from zero, including them in your model would probably be a good thing. So, in your terms, both for ANOVA(reduced model, full model) and for ANOVA(full model, reduced model) goes:

- Null: removed factors are not different from 0, full model is not better
- Alt: removed factors are different from 0, full model is better
- $p < 0.05 \Rightarrow$ choose full model
- $p > 0.05 \Rightarrow$ choose reduced model

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answered Mar 1, 2018 at 15:27



hj.linssen

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Assuming your models are nested, which I think is safe given that you have referred to them as full and reduced models, we'll consider the first model. As stated in the helptext for anova.lm, "It is conventional to list the models from smallest to largest, but this is up to the user."

In this case: **ANOVA(reduced model, full model)**, the null hypothesis is that the coefficients for all variables in the full model that are not in the reduced model are zero. The alternative hypothesis is that those coefficients are not zero.

If your p-value (what kind of test are you doing?) is < 0.05 you could infer that the data supports the alternative hypothesis that the coefficients *not shared between the models* are not zero.

Be aware, though, that before you make any assumptions about the meaning of the analysis, you need to be reasonably sure that the data meets the assumptions of the ANOVA and those involved in the calculation of p-values ([ANOVA assumption normality/normal distribution of residuals](#))

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edited Apr 13, 2017 at 12:44



Community Bot

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answered Apr 4, 2017 at 19:40



Eumenedies

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